Understanding Bird Breeding in NSW

Five questions to help prevent species decline

Client briefing · CS-13 Combined Breeding Analysis team · 17 October 2026

Data: NSW Bird Atlas (1977–2020) and DES backlog (to 2025)

THE FIVE QUESTIONS

Q1 **Which species are being seen less often and breeding less when they are seen?**

12 species are seen less often and breed less when seen — a watchlist for follow-up (named on the Question 1 page).

Q2 **Where in NSW has breeding activity faded the most, and where is it holding up?**

Breeding faded more than the NSW trend in 64% of inland and 61% of tablelands areas, vs 21% on the coast; 56 areas held up and may act as refuges.

Q3 **When in the year is breeding at its busiest, and when is the quiet window for disturbance?**

Most species nest Oct–Jan. Mar–Jul is the quiet window for planned disturbance such as burns or clearing.

Q4 **Are birds of the inland faring worse than coastal and tablelands birds?**

Inland birds are the group most affected: 21% show a big breeding drop vs 10% of coastal birds.

Q5 **Are we still monitoring enough to notice a species in decline?**

Survey effort is down about 92% on the 1977–1999 average, with only ~38 active observers a year.

HOW TO READ THIS REPORT

Each page asks one research question, answers it with a chart, and explains what it means for protecting species. Because the way breeding was recorded changed over time, species and places are compared with the typical NSW trend, not with raw totals.

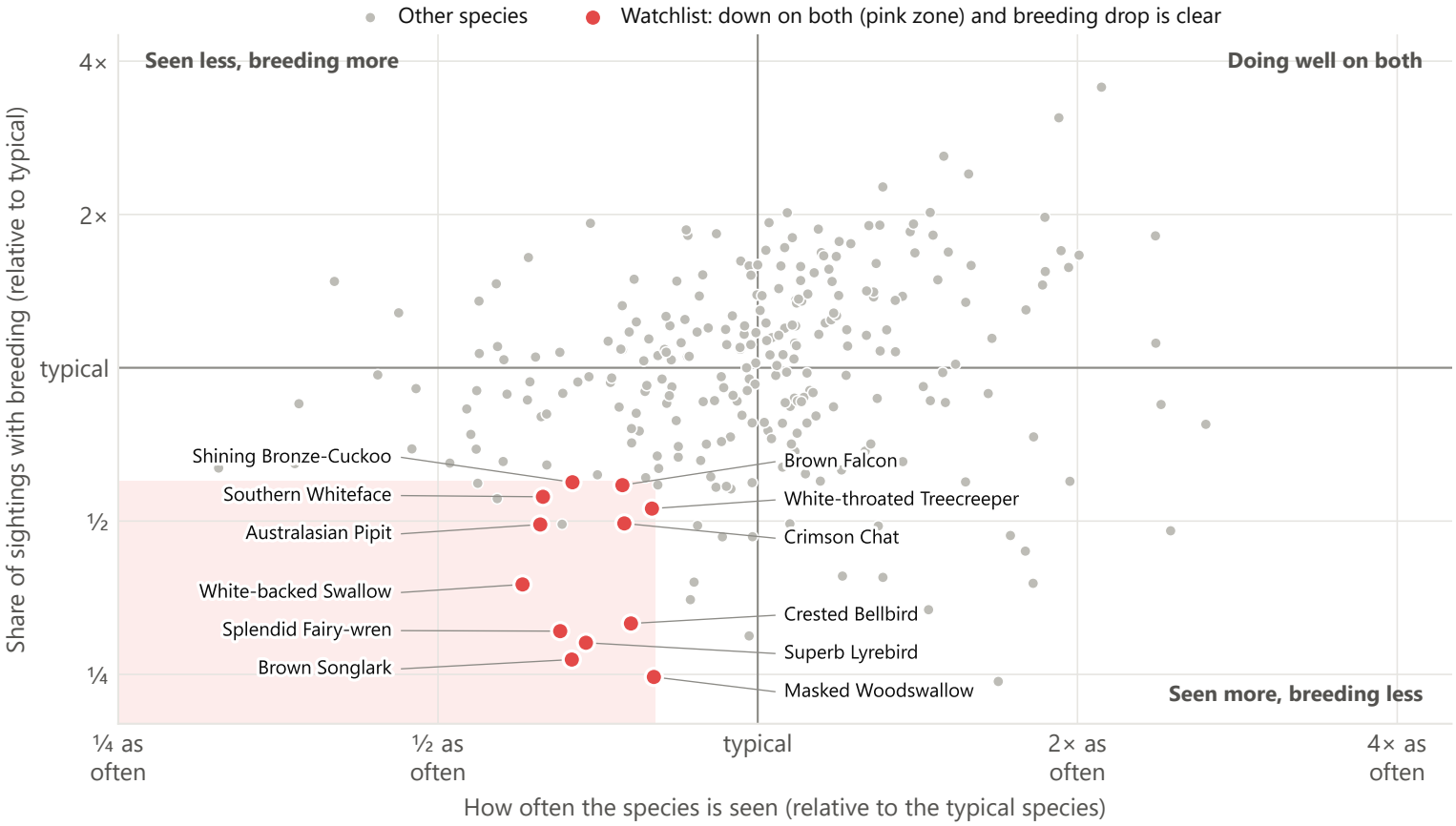
RESEARCH QUESTION 1

Which species are being seen less often and breeding less when they are seen?

Why this matters: A species still being seen but no longer breeding is an early warning: adults can live for years after breeding fails. Species down on both counts are strong candidates for a conservation watchlist.

12 species are seen less often AND breed less when seen

Each dot is one species, 2000–2020 compared with 1977–1999, relative to the typical species (1 = typical)



ANSWER

- Of 273 species with enough records, 83 fall below typical on both measures, but most of these are small differences.
- 12 species stand out: seen at least 20% less often than a typical bird AND breeding share at least 40% lower, with a breeding drop unlikely to be chance.
- The steepest breeding falls are Masked Woodswallow, Brown Songlark, Superb Lyrebird.
- 9 of the 12 are mainly inland birds (Masked Woodswallow, Brown Songlark, Splendid Fairy-wren, Crested Bellbird, White-backed Swallow and others) — Q4 tests whether inland birds as a group are faring worse.

What it means: These species are the best candidates for targeted follow-up: nest monitoring, habitat checks in their remaining strongholds, and comparison with official threatened-species listings.

Caution: Some watchlist species have few recent breeding records, so the size of each change is uncertain. Results are associations, not proof of cause.

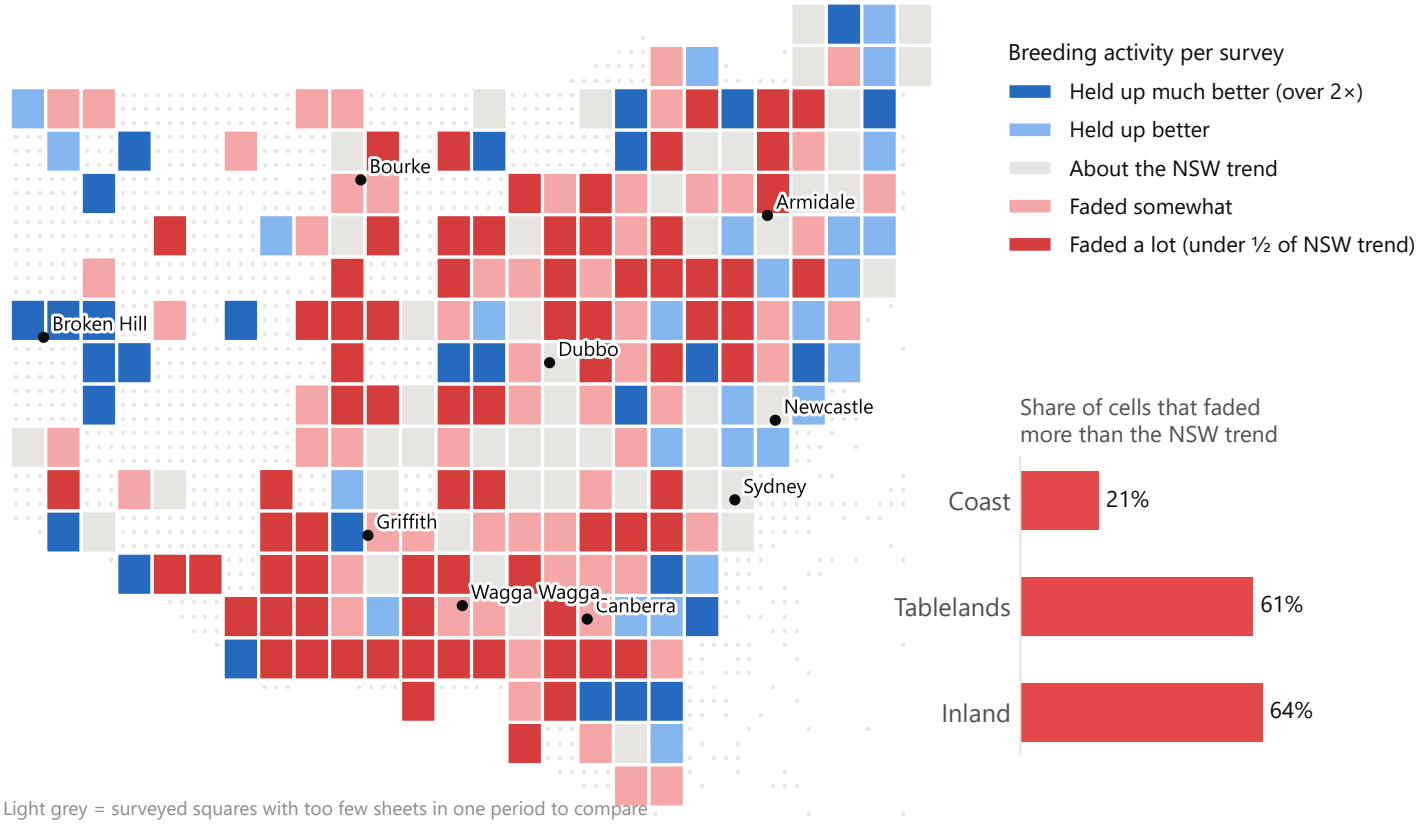
RESEARCH QUESTION 2

Where in NSW has breeding activity faded the most, and where is it holding up?

Why this matters: Protection is applied to places. Knowing where breeding has faded, and where it is still strong, shows where to look first and which areas may act as refuges.

Breeding faded in 64% of inland and 61% of tablelands areas, but only 21% on the coast

Change in breeding records per survey, 2000–2020 vs 1977–1999, compared with the NSW-wide trend



Light grey = surveyed squares with too few sheets in one period to compare

ANSWER

- Across 241 well-surveyed cells, 57% show breeding activity fading faster than the NSW-wide trend.
- By zone: 21% of coastal cells, 61% of tablelands cells and 64% of inland cells faded more than the state trend.
- 56 cells (23%) held up better than the state trend — these are candidate breeding strongholds worth protecting.
- Large parts of the far west have too few surveys in one or both periods to compare (light grey dots).

What it means: Inland and tablelands areas with faded breeding are priority areas to check on the ground; cells that held up are candidate refuges where protecting habitat may safeguard breeding.

Caution: Cells mix many species and observers changed over time; a faded cell is a prompt to investigate, not proof of habitat loss.

RESEARCH QUESTION 3

When in the year is breeding at its busiest, and when is the quiet window for disturbance?

Why this matters: Activities that disturb nests — hazard-reduction burns, land clearing, tree lopping, construction — can often be timed. Knowing when most species nest helps set safer windows.

Most species nest Oct–Jan; Mar–Jul is the quiet window

Share of each zone's breeding species that are actively breeding in each month (Atlas 2000–2020)

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
Coast (151 species)	14%	34%	**58%**	**71%**	**78%**	**64%**	20%	10%	5%	3%	2%	5%
Tablelands (104 species)	11%	33%	**68%**	**79%**	**72%**	**60%**	36%	5%	1%	2%	1%	1%
Inland (121 species)	21%	34%	**55%**	**75%**	**79%**	44%	26%	3%	1%	0%	1%	5%

Busy season — avoid disturbance near nests **Quiet window — lowest risk**

A month counts as active for a species when its breeding rate is at least 50% of its peak month.

ANSWER

- Breeding is strongly seasonal everywhere: at least half of a zone's species are active from Oct to Jan, peaking in Dec (79% of species in the busiest zone).
- Mar to Jul is the quiet window: fewer than 10% of species are breeding in any zone.
- The season ends earlier inland: in January 44% of inland species are still active, versus 64% on the coast and 60% on the tablelands.
- The build-up starts in August: by September about 34% of species are already breeding.

What it means: Scheduling planned disturbance in Mar–Jul, and avoiding Oct–Jan, would protect the most nests. Early breeders (such as some owls and honeyeaters) still need site-specific checks.

Caution: This is the pattern across many species; individual threatened species can breed outside it.

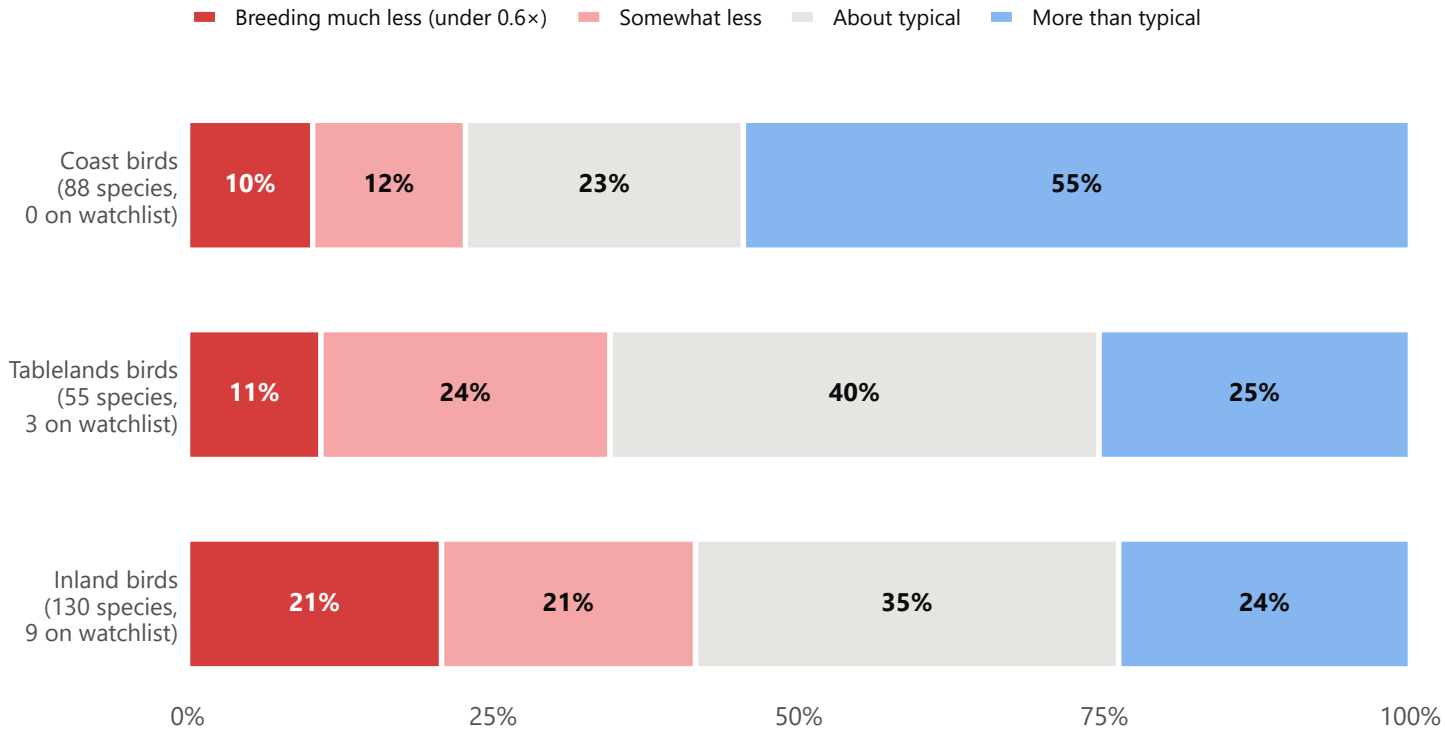
RESEARCH QUESTION 4

Are birds of the inland faring worse than coastal and tablelands birds?

Why this matters: If declines are concentrated in one group of birds, it points to a shared pressure and suggests protecting that group's habitat as a whole rather than one species at a time.

Inland birds are 2.0× as likely as coastal birds to show a big breeding drop

Share of species in each home zone by change in breeding share (2000–2020 vs 1977–1999)



Home zone = the zone where the species was reported most often in 1977–1999. 'Typical' = changed about the same as most species.

ANSWER

- 21% of inland species show a big breeding drop, compared with 11% of tablelands species and 10% of coastal species.
- Counting any drop below typical: inland 42%, tablelands 35%, coast 23%.
- 9 of the 12 watchlist species from Q1 are inland birds.
- Coastal birds mostly held up: 55% bred more than the typical species.

What it means: Inland woodland and open-country birds are the group most in need of attention. Protecting and restoring inland woodland habitat could benefit several at-risk species at once.

Caution: Home zone is a simple label; many species occur in more than one zone. Weather and drought data have not yet been supplied, so drought effects cannot be separated out.

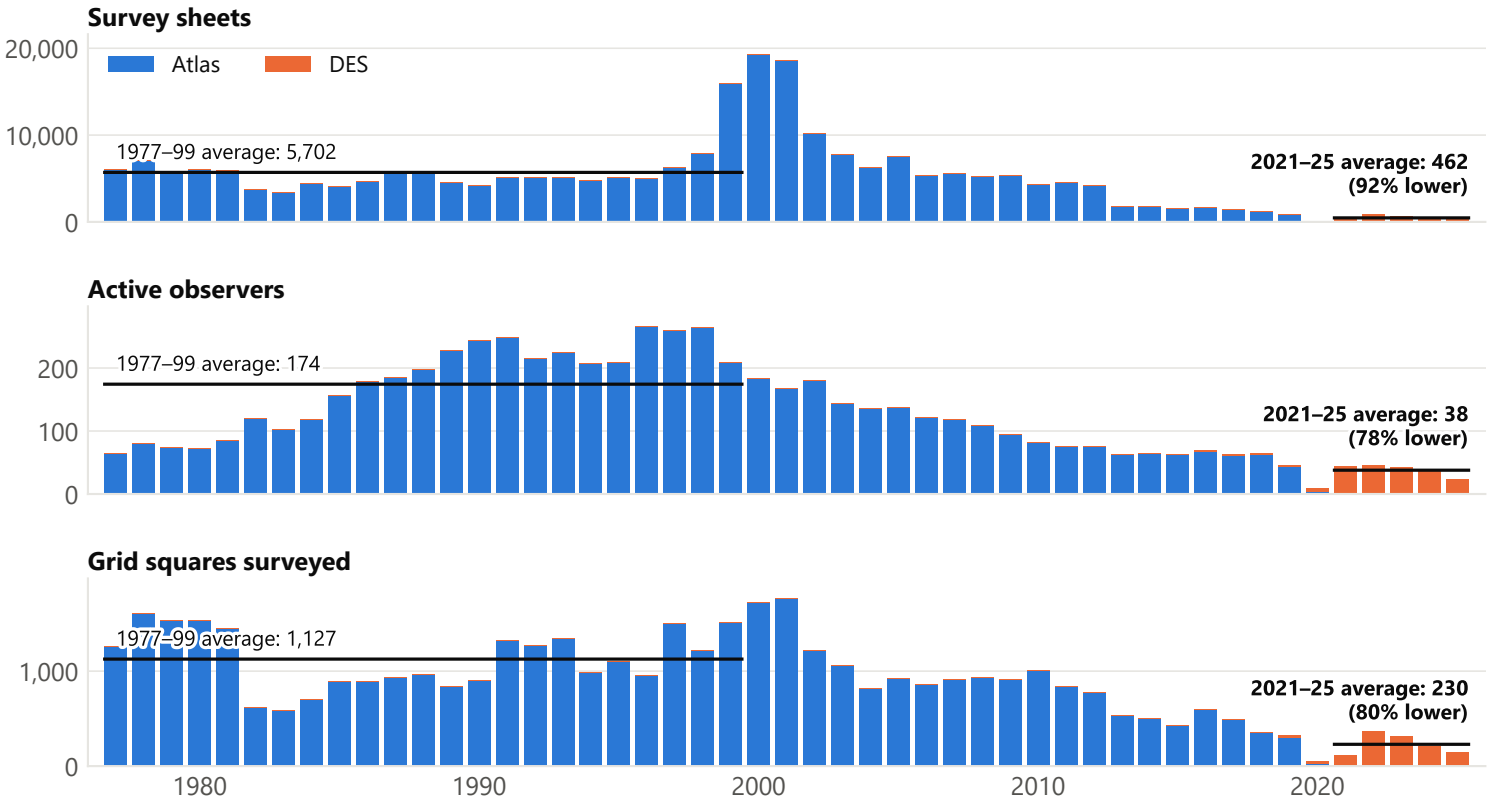
RESEARCH QUESTION 5

Are we still monitoring enough to notice a species in decline?

Why this matters: Every other answer depends on people surveying. If monitoring thins out, a species can decline without anyone noticing — the earlier warning signs in Q1 to Q4 would simply go unseen.

Survey effort is down about 92% on the 1977–1999 average

Survey effort per year, Atlas and DES (2020 is a partial year: the Atlas ends in March 2020)



ANSWER

- Survey sheets averaged 5,702 a year in 1977–1999 but only 462 a year in 2021–2025 — about 92% lower.
- Active observers fell from about 174 a year to about 38 (78% lower).
- Grid squares surveyed fell from about 1,127 a year to about 230 (80% lower), so much of NSW now goes unvisited in a typical year.
- The recent record leans on a few people: the top 5 observers supplied 37% of survey sheets in 2021–2025.

What it means: Current monitoring is too thin to track most species reliably. Rebuilding a volunteer network, with repeat visits to the watchlist species' strongholds and inland cells, is likely the single most important step for early detection.

Caution: DES is a different recording system from the Atlas; here it is only used to count survey effort.

SUMMARY

What this means for protecting bird species

Who	12 species are seen less often and breed less when seen — a watchlist for follow-up (named on the Question 1 page).
Where	Breeding faded more than the NSW trend in 64% of inland and 61% of tablelands areas, vs 21% on the coast; 56 areas held up and may act as refuges.
When	Most species nest Oct–Jan. Mar–Jul is the quiet window for planned disturbance such as burns or clearing.
Which group	Inland birds are the group most affected: 21% show a big breeding drop vs 10% of coastal birds.
Can we see it	Survey effort is down about 92% on the 1977–1999 average, with only ~38 active observers a year.

SUGGESTED NEXT STEPS

1. Check the watchlist against official threatened-species listings and prioritise species not yet listed.
2. Target new surveys (including nest checks) at the faded inland cells and at the strongholds that held up.
3. Recommend timing planned burns and clearing for Mar–Jul where practical.
4. Rebuild the volunteer monitoring network, with repeat visits to the same sites each spring.
5. Add rainfall/drought and land-cover data so that possible causes can be tested.

ABOUT THE DATA AND METHOD

- Records: 6,634,025 usable species records from the NSW Bird Atlas and the DES backlog. eBird is excluded, as agreed with the client.
- Q1–Q4 compare the Atlas with itself, 1977–1999 vs 2000–2020, using surveys only. Atlas and DES rates are never mixed because they recorded breeding differently.
- Breeding was recorded less often overall in recent years, so each species and place is compared with the typical change rather than raw totals.
- Q1 includes species with at least 300 sightings in each period and 30 early breeding records. A drop is 'clear' when it is unlikely to be chance after allowing for testing many species.
- All settings (periods, thresholds, map cell size) can be changed in the accompanying notebook.